Comparison Operators

The Logic Behind Decision Making

Unlike arithmetic operators that return numbers, **Comparison Operators** compare two values and return a **Boolean** result:

true or **false**

1. Quick Reference Table

Operator	Name	Logic Check
`==`	Equal to	Is x exactly equal to y?
`!=`	Not equal	Is x different from y?
`>`	Greater than	Is x larger than y?
`<`	Less than	Is x smaller than y?
`>=`	Greater or Equal	Is x larger or the same as y?
`<=`	Less or Equal	Is x smaller or the same as y?

2. Decision Logic: Voting Eligibility

Comparison operators are the engine inside "If-Statements". They help the computer decide if code should run or not.

```java
int age = 18;

// Check if age is 18 or older
System.out.println(age >= 18); // Output: true

// Check if age is under 18
System.out.println(age < 18); // Output: false
```

3. The "Not Equal" Operator

A common operator used to verify that a value has changed or is not a specific number.

```java
int status = 404;
System.out.println(status != 200); // Output: true (404 is NOT equal to 200)
```

Key Takeaways

- Comparison results are always **boolean** values.
- **==** is for comparing; **=** is for assigning. Don't mix them up!
- These operators are essential for **Conditionals** (Loops and If-statements).
- They work with both primitives (int, double) and chars.